GEOGRAPHY

PHYSICAL GEOGRAPHY

MODULE - 14

- PLATE TECTONICS
- DIVERGENT BOUNDARY
- CONVERGENT BOUNDARY
- TRANSFORM BOUNDARY

Learning Space

HOW THE LITHOSPHERE INTERACTS WITH OTHER SPHERES?

These spheres interact to influence such diverse elements as salinity, biodiversity, landscape.

1. We have already seen that Lithosphere is cooler and brittle.
2. It is one of the five great spheres that shape the environment of the Earth.
3. The other four are ...
   - Biosphere, which looks at the Earth's living things.
   - Cryosphere, which looks at Earth's frozen regions, including Ice and frozen soil.
   - Hydrosphere, which looks at Earth's liquid water.
   - Atmosphere, which looks at the air surrounding our Planet.

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WHAT ARE THE FORCES BEHIND PLATE TECTONICS?

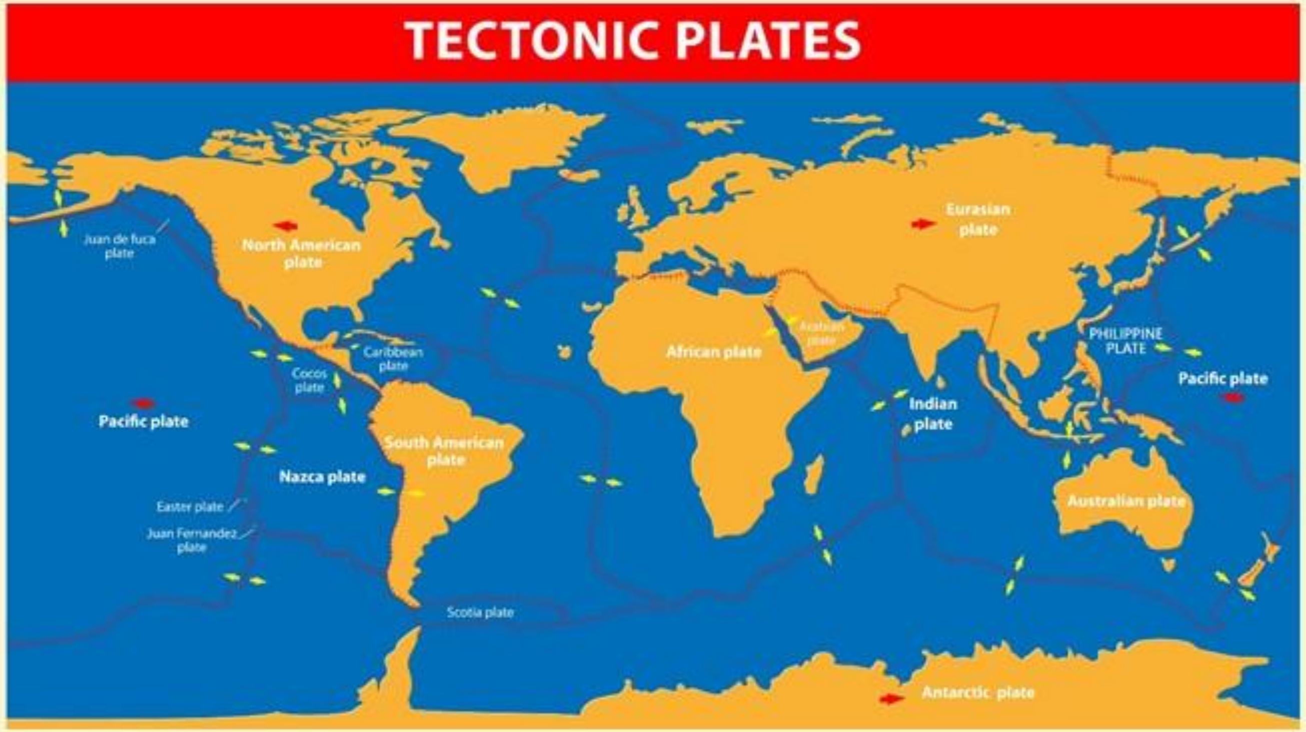

Plate Tectonics is driven by variety of forces, some of them are...

- Dynamic movement in the Mantle.
- Dense Oceanic Crust interacting with the ductile asthenosphere.
- Rotation of the Planet.

PLATE TECTONICS

It is the movement and interaction of the Earth's Plates. The plates move at the rate of 3-5 cm per year.

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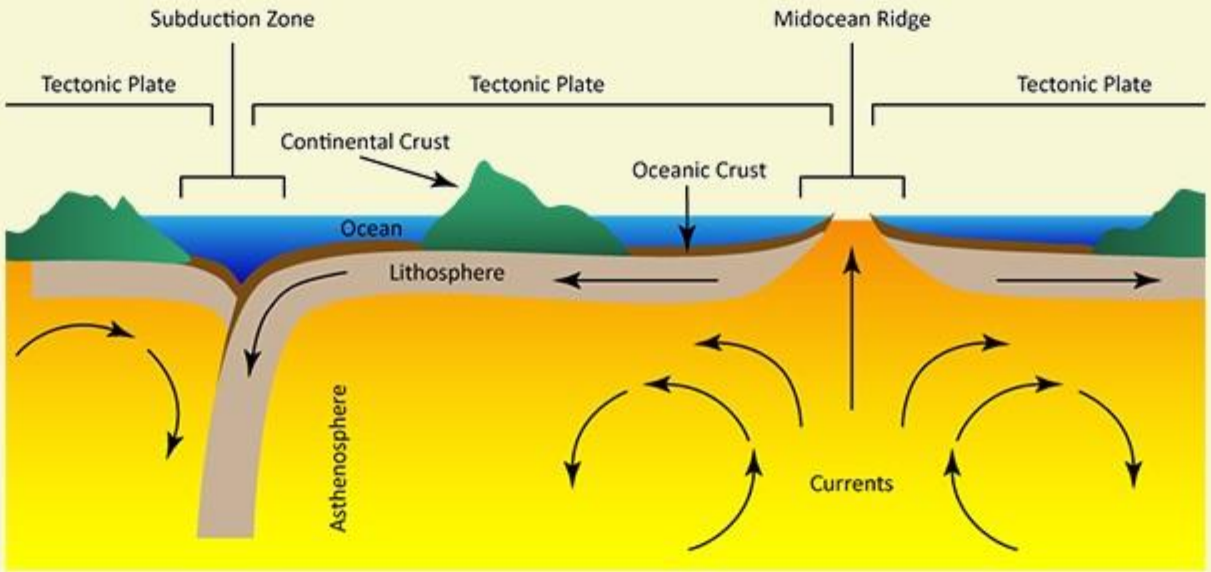
PLATE BOUNDARIES AND THEIR INTERACTIONS



1. We have already seen Lithosphere is comprised of various plates.
2. Heat raising and falling inside the Mantle creates convection currents generated by radioactive decay in the core.
3. The convection currents move the plates. When the convection currents diverge near the Earth's crust, plates move a part. When the convection currents converge, plates move towards each other.
4. The movement of the plates and the activity inside the Earth is known as Plate Tectonics and Plate Tectonics causes Earthquakes and the Volcanoes.

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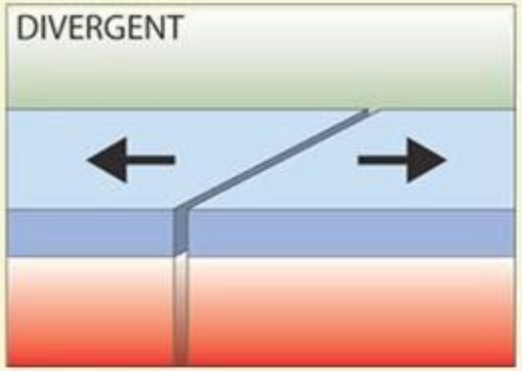
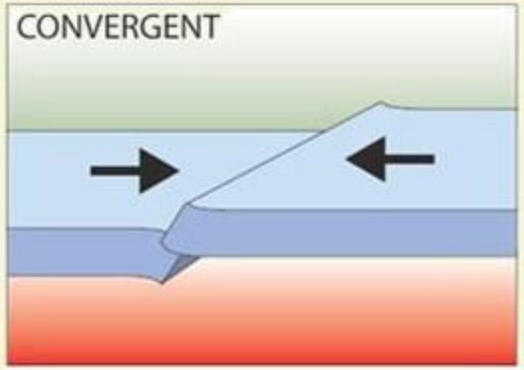
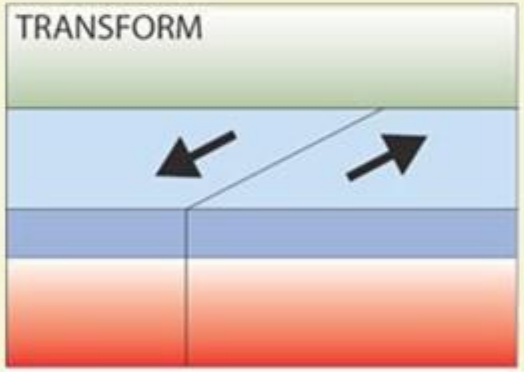
WHAT ARE DIFFERENT TYPES OF PLATE BOUNDARIES?







1. At a tensional, constructive or divergent boundary, the plates move apart.
2. At a compressional, destructive or convergent boundary, the plates move towards each other.
3. At a conservative or transform boundary, the plates slide past each other.

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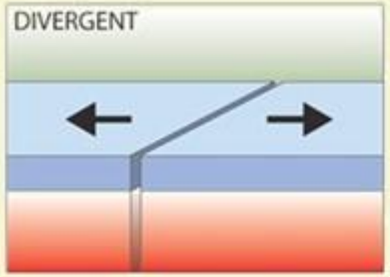
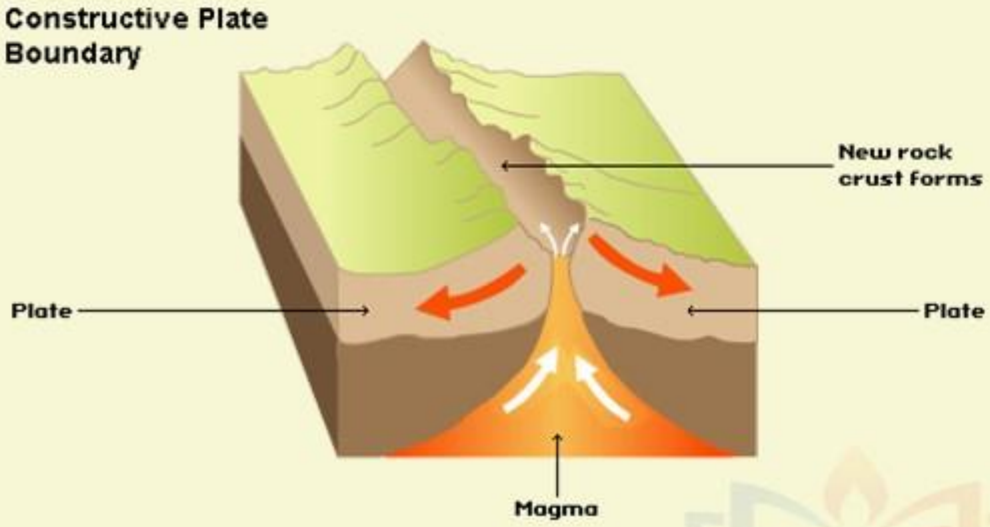
PLATE TECTONICS

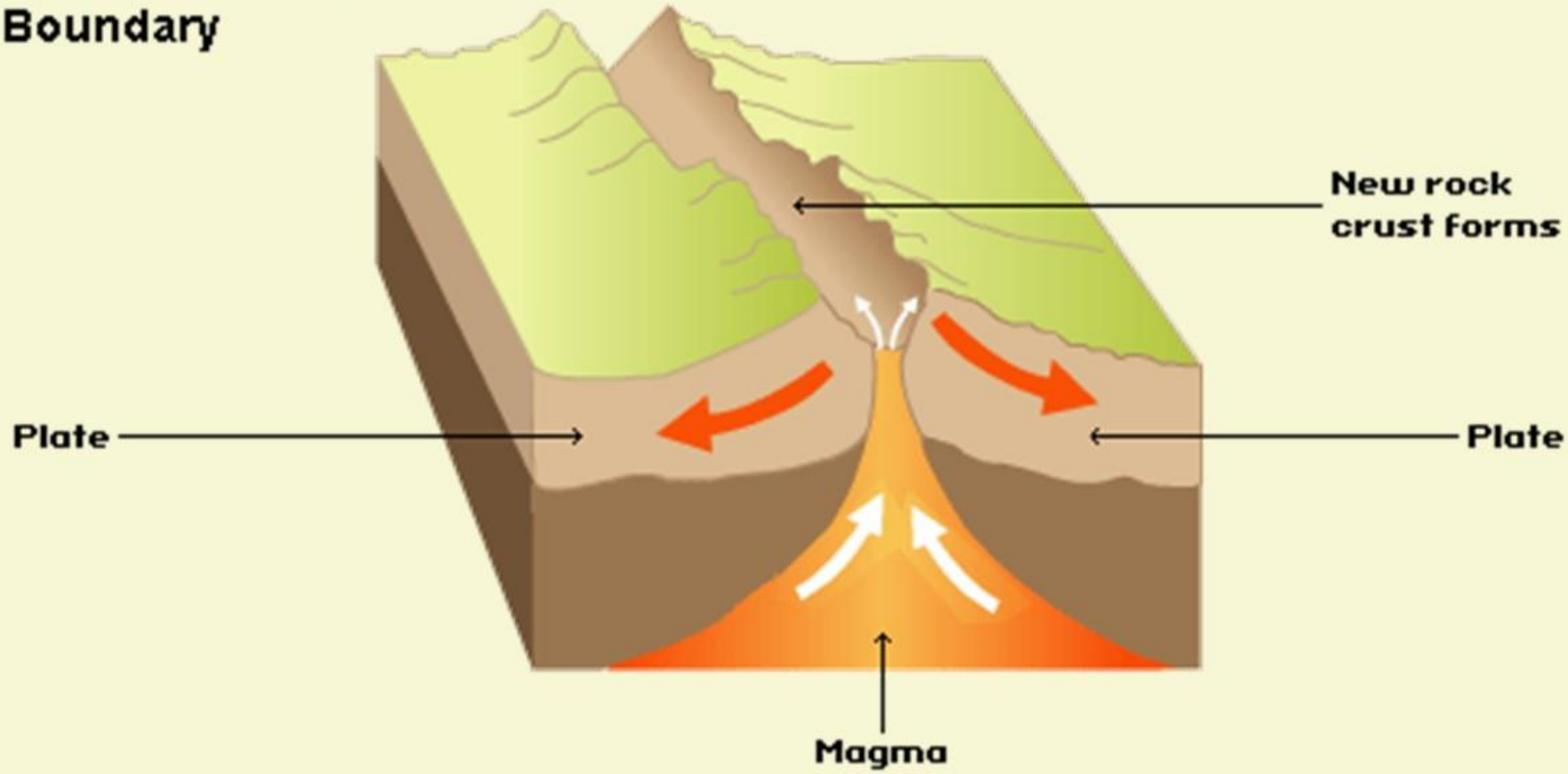
1 CONSTRUCTIVE / DIVERGENT BOUNDARY



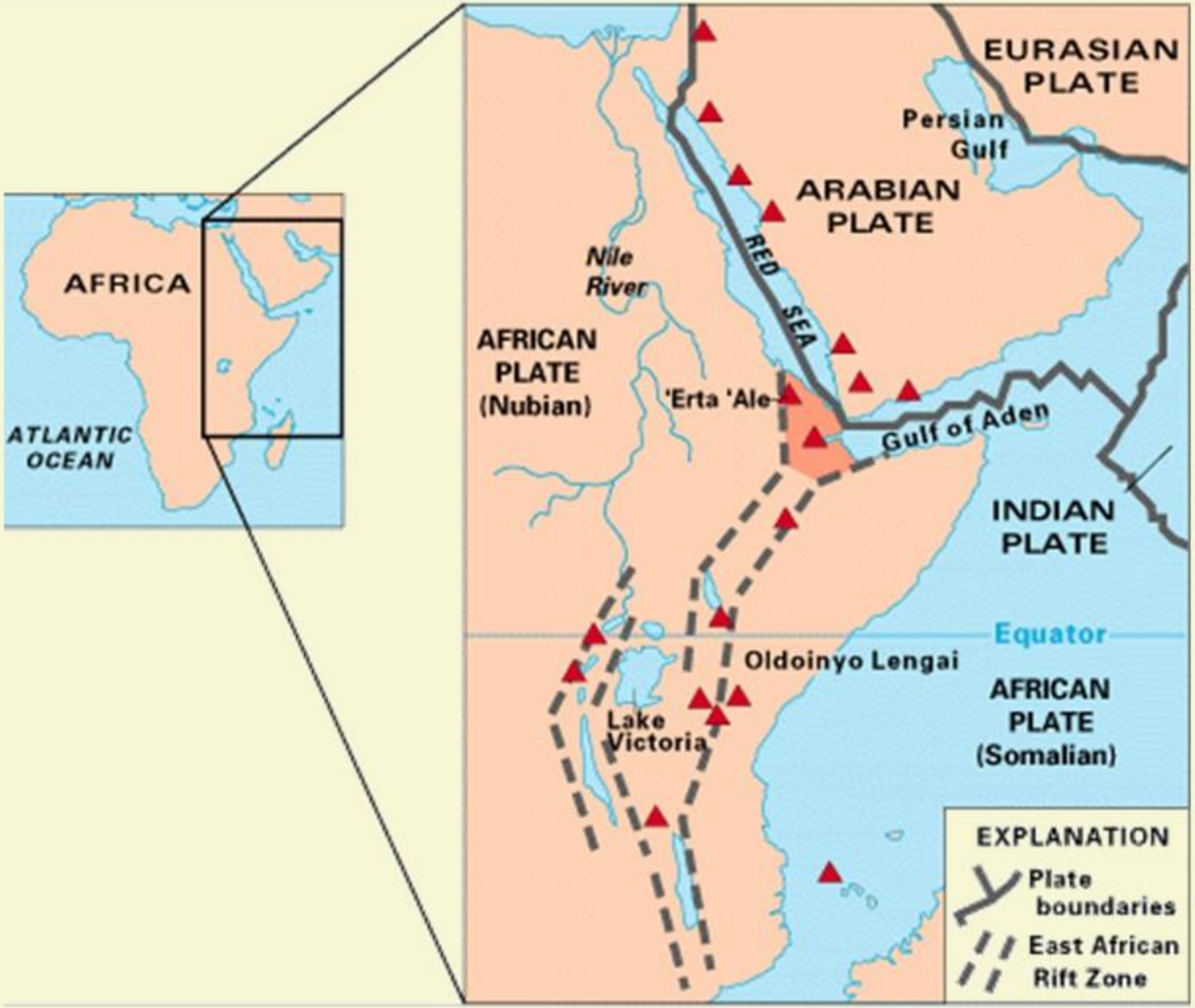


1. At divergent boundaries in the oceans, magma from deep in the Earth's mantle rises towards the surface and pushes apart two or more plates.
2. Mountains and volcanoes rise along the seam.
3. The process renews the ocean floor and widens the giant basins.
4. A single mid-ocean ridge system connects the world's oceans, making the ridge the longest mountain range in the world.
5. One example is giant troughs, such as the Great Rift Valley in Africa, which was formed, where plates are moving apart.

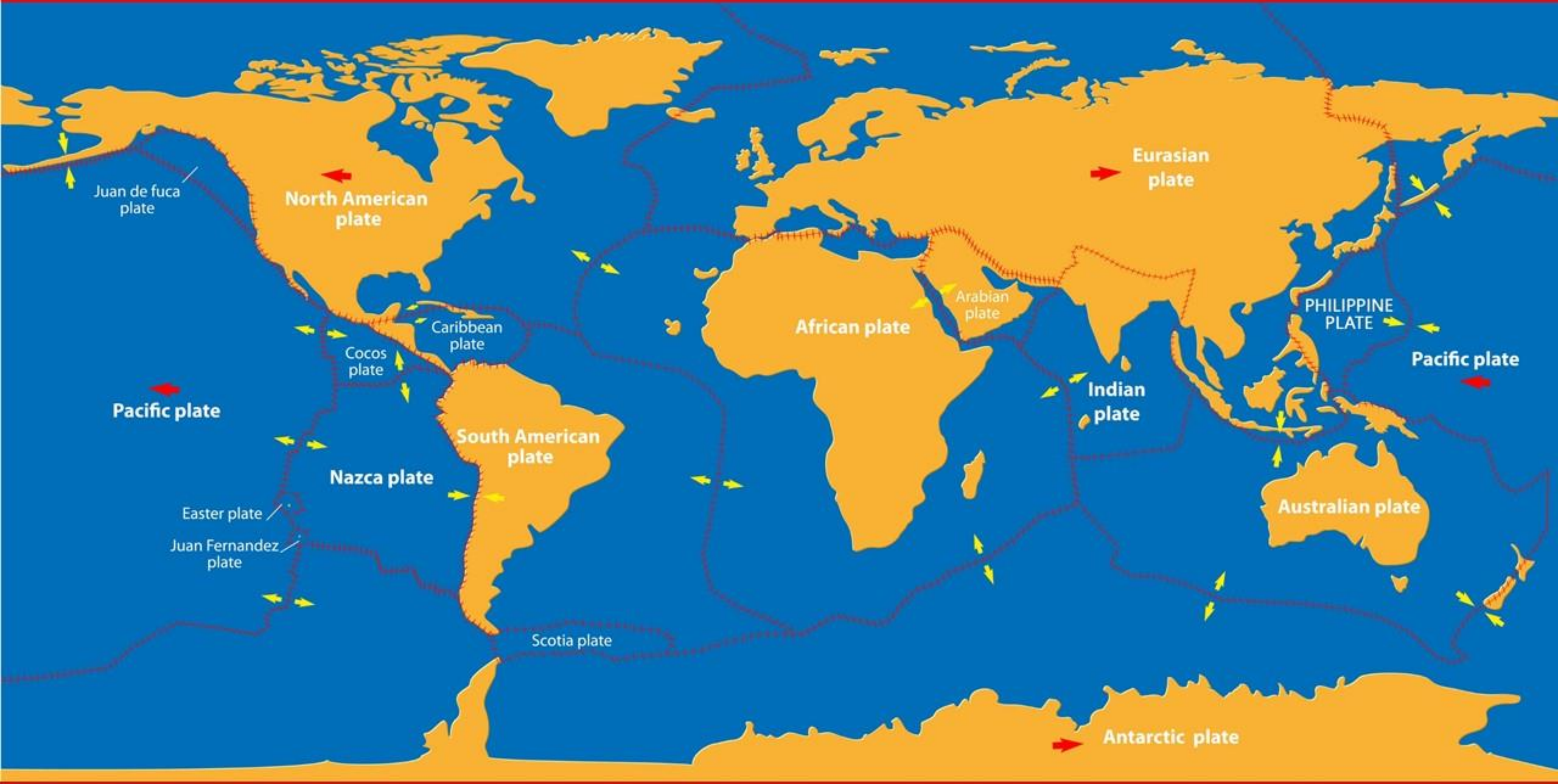
Constructive Plate Boundary

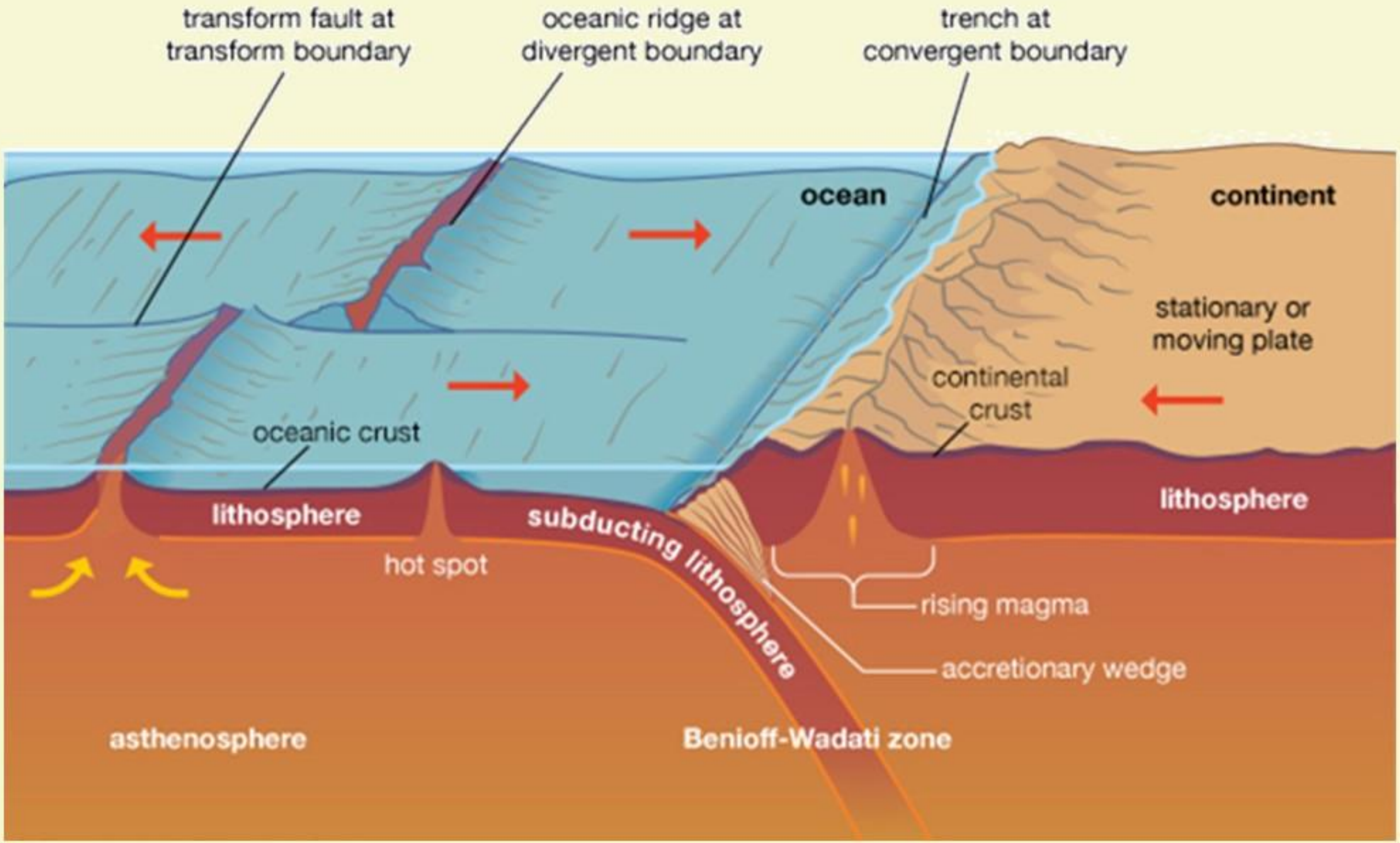


RIFT VALLEY
AFRICA



TECTONIC PLATES





transform fault at transform boundary
oceanic ridge at divergent boundary
trench at convergent boundary
ocean
continent
stationary or moving plate
continental crust
oceanic crust
lithosphere
subducting lithosphere
lithosphere
hot spot
rising magma
accretionary wedge
asthenosphere
Benioff-Wadati zone

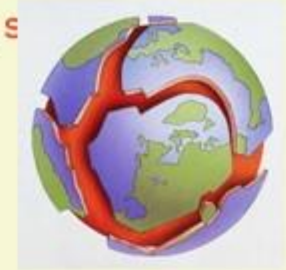
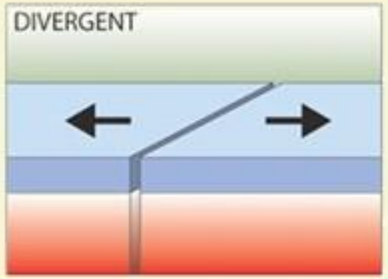


PLATE TECTONICS

1 MORE ON CONSTRUCTIVE / DIVERGENT BOUNDARY



6 The plates move apart due to the convection currents inside the Earth.

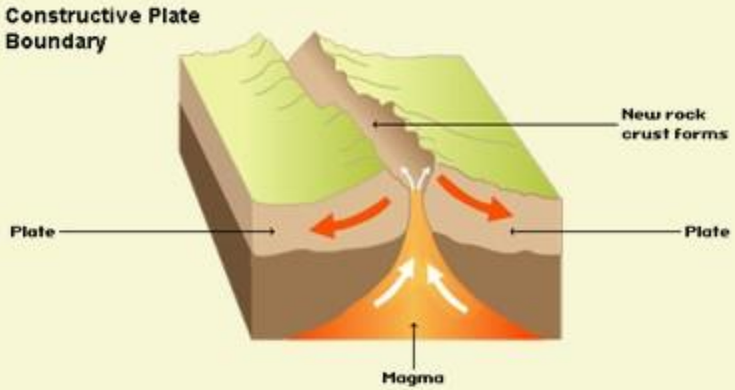
7 As the plates move apart (though slowly), magma raises from the Mantle. The magma erupts to the surface of the Earth. This is also accompanied by Earthquakes. When the magma reaches the surface, it cools and solidifies to form a new crust of igneous rocks.

8 This process is repeated several times over a long period of time.

9 Eventually, the new rock builds up to form a volcano.

11 We can see chains of underwater volcanoes formed along the plate boundary.

12 Some of these volcanoes become so large that they erupt out of the sea to form a volcanic island. Examples of volcanic islands are Surtsey and the Westman Islands near Iceland.




GREENLAND
ICELAND
SURTSEY ISLANDS
WESTMAN ISLANDS
CANADA
UNITED STATES
MID - ATLANTIC RIDGE
North Atlantic Ocean
MEXICO
VENEZUELA
COLOMBIA
SPAIN
FRANCE
GERMANY
POLAND
BELARUS
UKRAINE
ROMANIA
ITALY
TURKEY
Black Sea
Mediterranean Sea
SYRIA
IRAQ
IRAN
MOROCCO
ALGERIA
LIBYA
EGYPT
SAUDI ARABIA
Red Sea
MAURITANIA
MALI
Niger
CHAD
SUDAN
YEMEN
NIGERIA
ETHIOPIA
SOUTH SUDAN
SOMALIA
Cameroon
Gulf of Guinea
KENYA
DR CONGO
Learning Space

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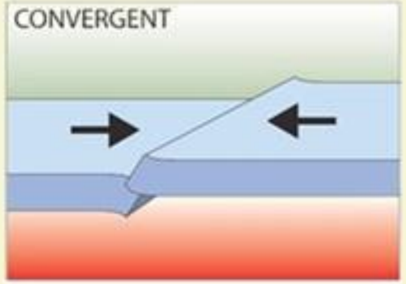
PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY



1. This is a compressional or destructive boundary. Here the plates move towards each other.
2. They usually involve a continental plate and an oceanic plate.
3. The oceanic plate is denser as we have already seen, than the continental plate.
4. The oceanic plate is forced underneath the continental plate.
5. The point, at which this happens is called the Subduction Zone.
6. As the oceanic plate is forced below the continental plate, it melts to form magma and earthquakes are triggered.
7. The magma collects to form a magma chamber.
8. The magma then rises up through the cracks in the continental crust.
9. When the pressure builds up, a volcanic eruption may occur.

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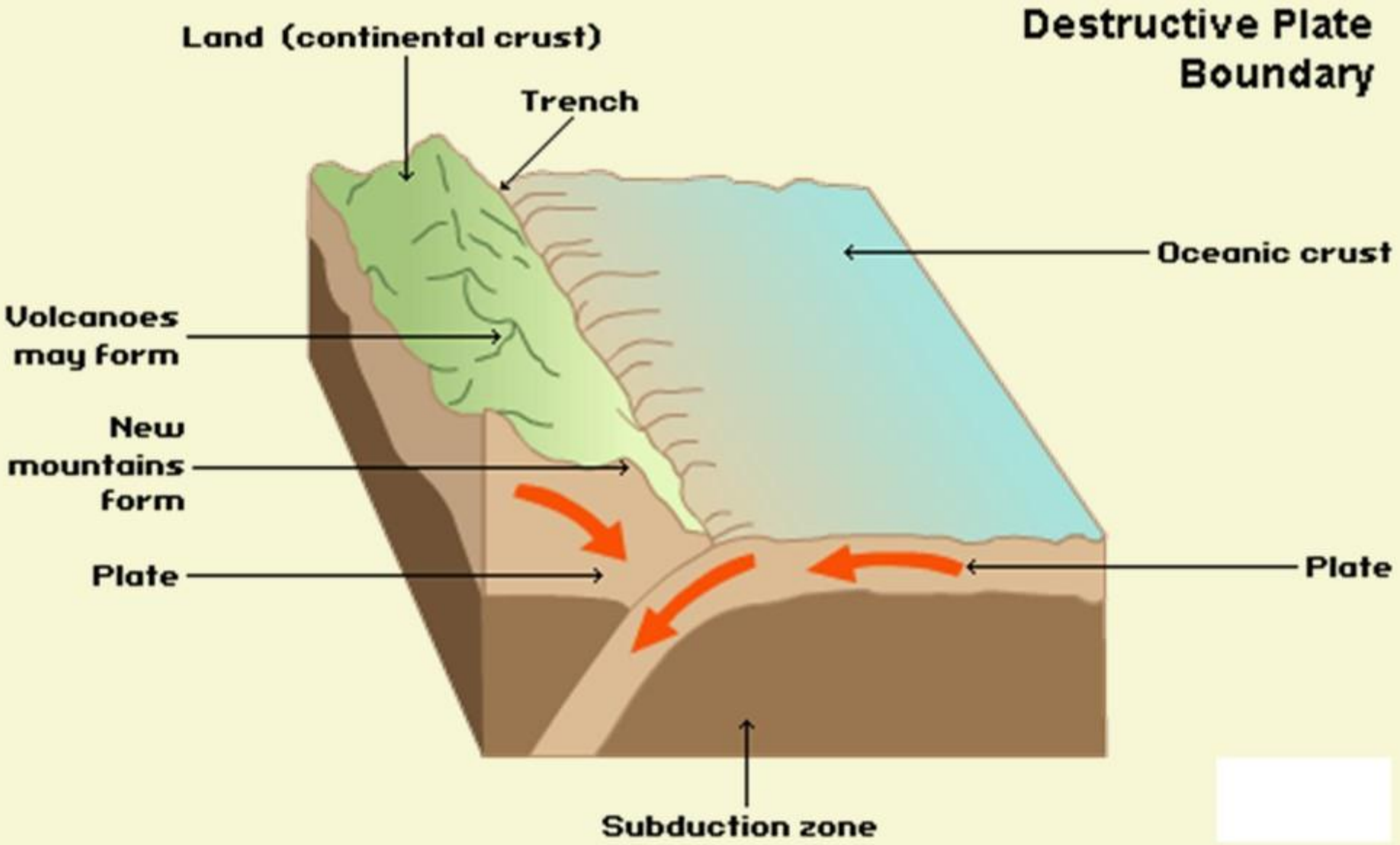
Destructive Plate Boundary
Land (continental crust)
Trench
Oceanic crust
Volcanoes may form
New mountains form
Plate
Plate
Subduction zone

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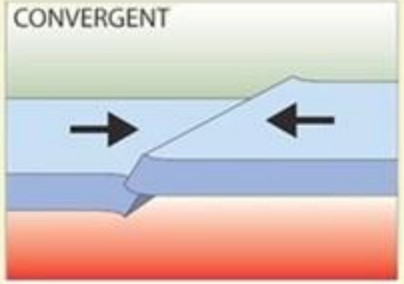
PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY



FOLD MOUNTAINS

1. As the plates push together, the continental crust is squashed together and forced upwards. This is called folding. The process of folding creates Fold Mountains.
2. When the plates serving the landmasses collide, the crust crumples and buckles into the mountain ranges.
3. India and Asia crashed about 55 million years ago, giving rise to the Himalayas, the highest mountain system on Earth.

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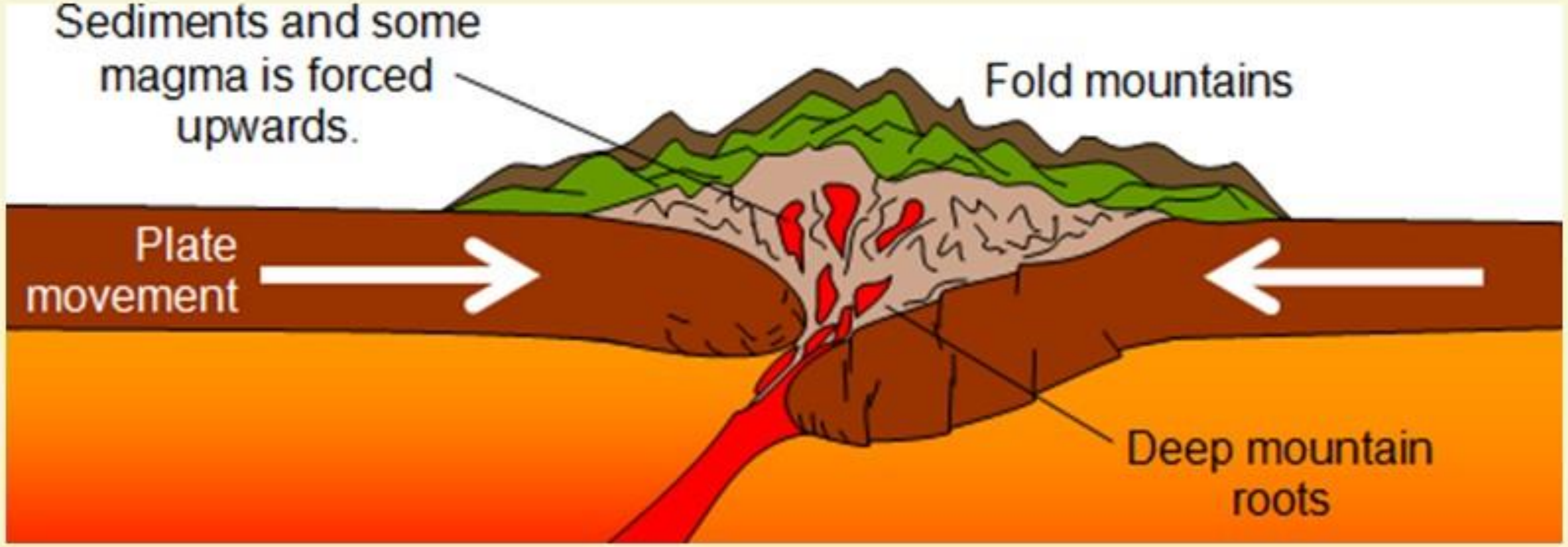
FOLD MOUNTAINS



TECTONIC PLATES

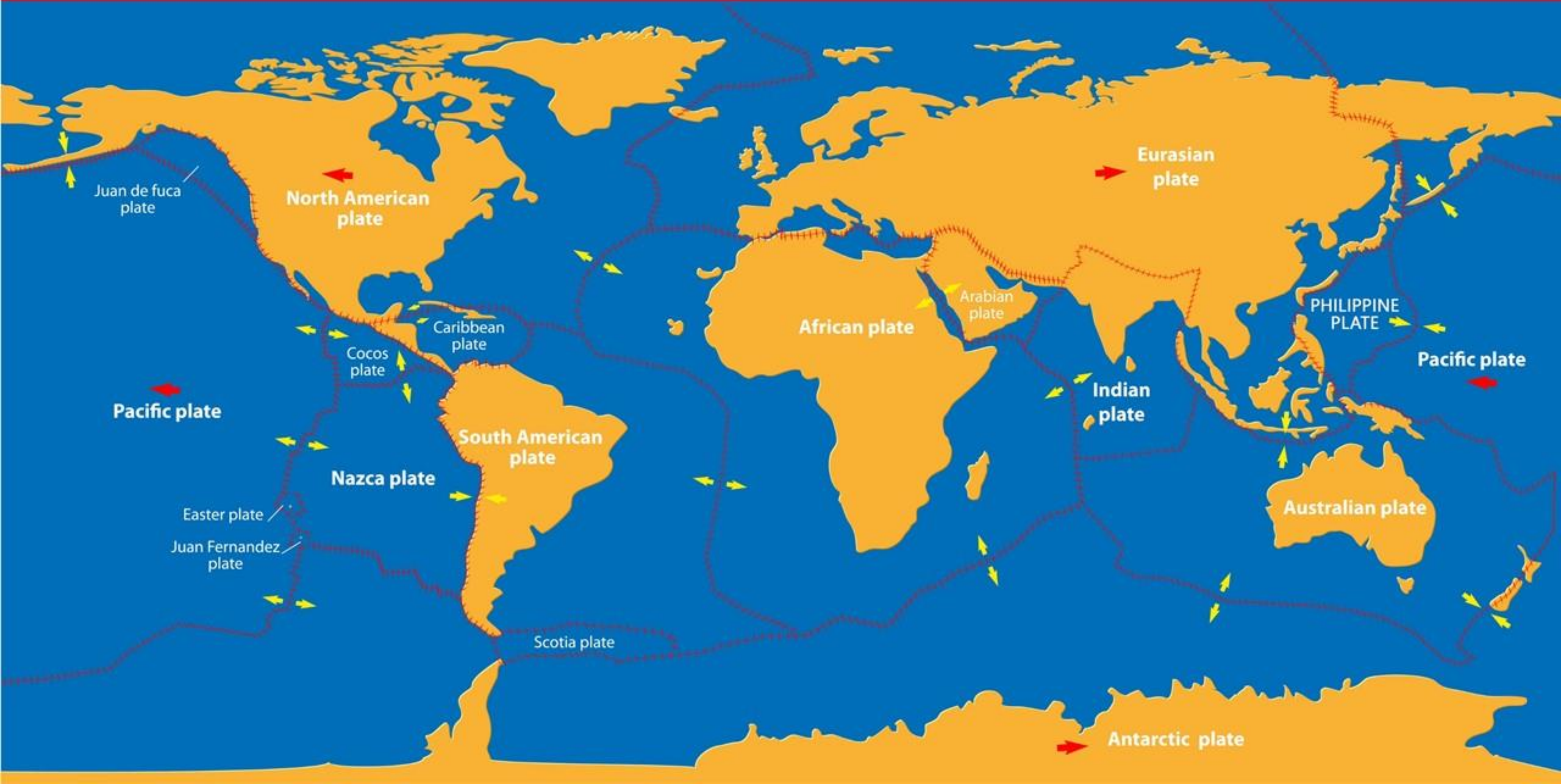
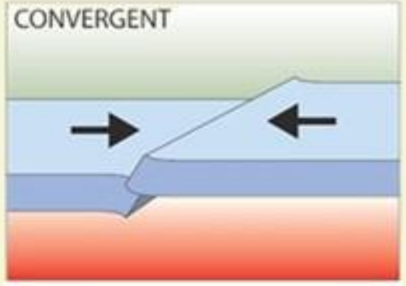
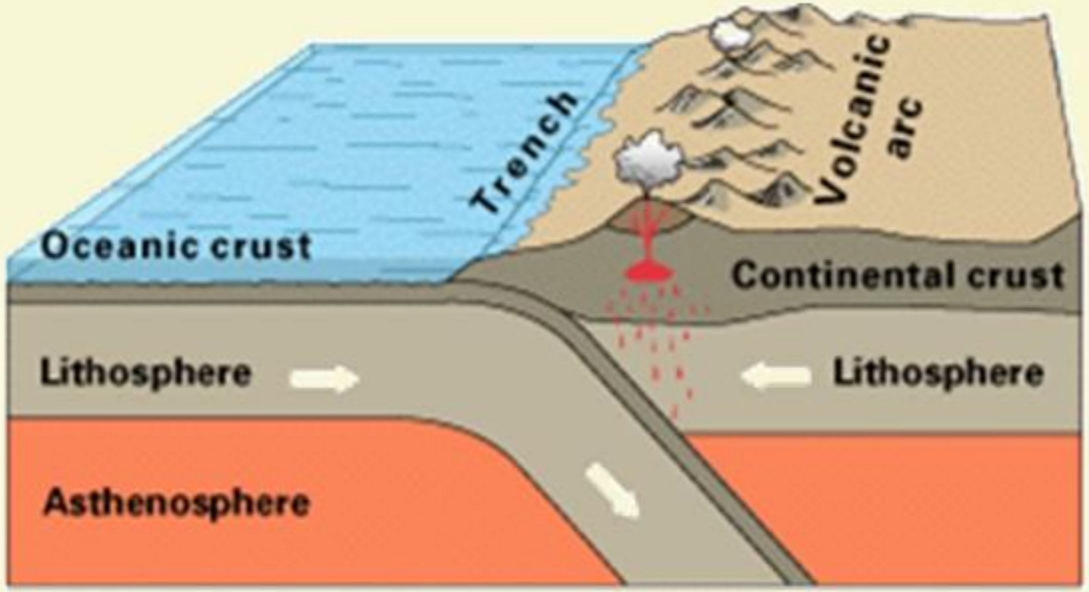


PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY





OCEANIC – CONTINENTAL CONVERGENCE

WHAT ABOUT ANDES MOUNTAINS?

1. Andes Mountains are formed because of the convergent boundary between Oceanic Plate and Continental Plate.
2. In addition, the diving plate melts and is often comes out in volcanic eruptions, the classic examples are those formed in the Mountains of Andes.

VENEZUELA
COLOMBIA
ECUADOR
PERU
BRAZIL
BOLIVIA
ANDES MOUNTAINS
PARAGUAY
CHILE
ARGENTINA
URUGUAY
South Pacific Ocean
South Atlantic Ocean
NIGER
Benin
IVORY COAST
Ghana
Gulf of Guinea

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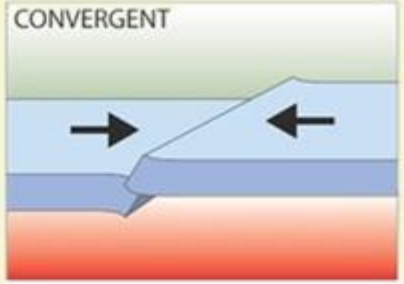
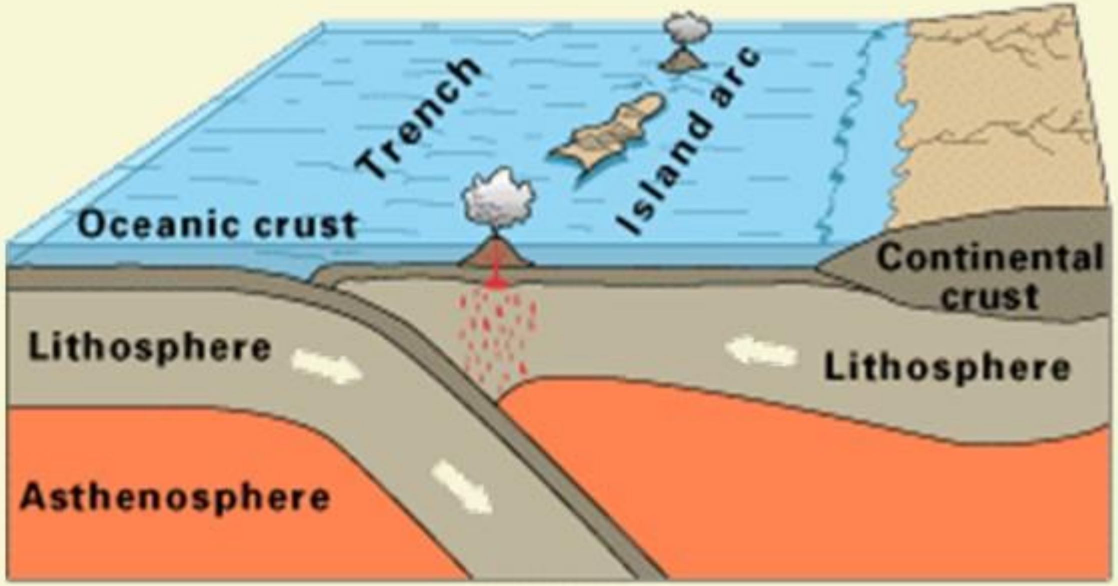
PLATE TECTONICS

2 DESTRUCTIVE / CONVERGENT BOUNDARY





OCEANIC – OCEANIC CONVERGENCE

OCEAN AND OCEAN CONVERGENCES

1. Because of the ocean-ocean convergences, one plate usually dives beneath the other, forming deep trenches like the Mariana Trench in the North Pacific Ocean, the deepest point on Earth.

2. These types of collisions can also lead to underwater volcanoes that eventually build up into island arcs like Japan.

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KYRGYZSTAN
TAJIKISTAN
CHINA
North Korea
South Korea
JAPAN
Sea of Japan
North Pacific Ocean
NEPAL
INDIA
BANGLADESH
MYANMAR (BURMA)
East China Sea
THAILAND
South China Sea
Philippine Sea
Mariana Trench
Bay of Bengal
Gulf of Thailand
Sea
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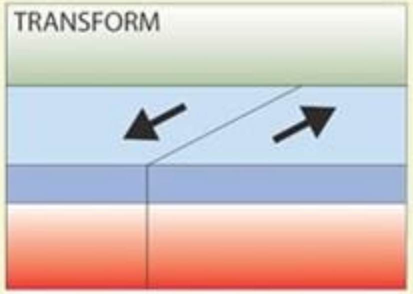
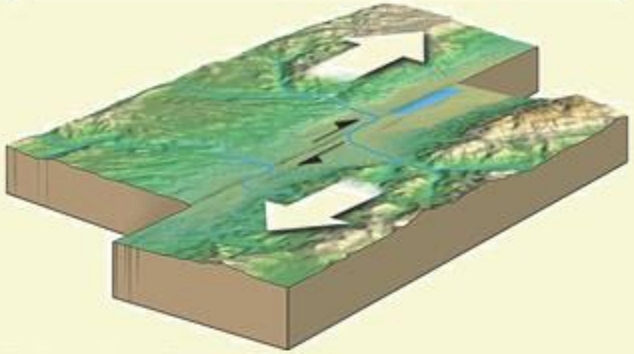
PLATE TECTONICS

3 CONSERVATIVE / TRANSFORM BOUNDARY





TRANSFORM BOUNDARIES

SAN ANDREAS FAULT IN CALIFORNIA

1. Here the two plates grind past each other, what are called strike-slip faults.
2. These boundaries do not produce spectacular features like mountains or oceans ridges, but the halting motion often triggers large earthquakes, such as the 1906 that devastated San Francisco.
3. The San Andreas Fault in California is an example of a transform boundary.

Learning Space


Point Reyes
San Francisco
Los Trancos
San Juan Bautista
Pinnacles NP
SAN ANDREAS FAULT
NORTH AMERICAN PLATE
Parkfield
PACIFIC PLATE
Carrizo Plain
Palm Springs

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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